BRSM Movie Memory Experiment

Based on Event Segmentation Theory · BRSM Project 2026 | IIIT

Team - MSN

Motivation

- Human experiences are not stored as one continuous stream.
- The brain divides experiences into **events**.
- These event boundaries help structure memory.

Key Question

What happens to memory when natural event boundaries are disrupted?

Event Segmentation Theory (EST)

Event Segmentation Theory explains how humans process continuous experiences.

Key Ideas

- The brain divides experiences into **discrete events**
- Boundaries occur when **prediction errors happen**
- Changes in environment trigger boundaries

Examples of Cues

- Camera angle change
- Lighting change
- Location change

Research Objective

The goal of this study is to examine how **disrupting event boundaries** affects memory.

We compare two conditions:

Natural Cut Videos	Abrupt Cut Videos

Memory performance is measured using:

- Recognition Accuracy
- Response Time
- Confidence Ratings

Research Question

> **How do abrupt cuts vs natural cuts in videos affect memory recognition?**

Abrupt cuts may:

Disrupt Event Model Updating

Abrupt cuts interrupt the brain's ongoing prediction process

Interrupt Event Continuity

The natural flow of an event is broken before completion

Impair Memory Encoding

Encoding of the event into long-term memory may be compromised

Experimental Design

Two-phase experiment

Phase	Description
Encoding	Participants watch 40 short videos
Recognition	Participants find previously seen frames (target vs lure)

Participants were divided into two groups:

Natural Cut group (NB)

- Videos ended at **natural event boundaries**

Abrupt Cut group (AB)

- Videos stopped **1–5 seconds before the boundary**

Vigilance check: 5 repeated videos — participants press spacebar on detection

Stimuli & Frame Types

Video Stimuli

- **40 short YouTube Shorts videos**
- Event boundaries found by annotators

Recognition test: **Target image** (from video) vs **Lure image** (similar, unseen)

Two Frame Categories

BB (Before Boundary)

Frames right before the event boundary

EM (Event Middle)

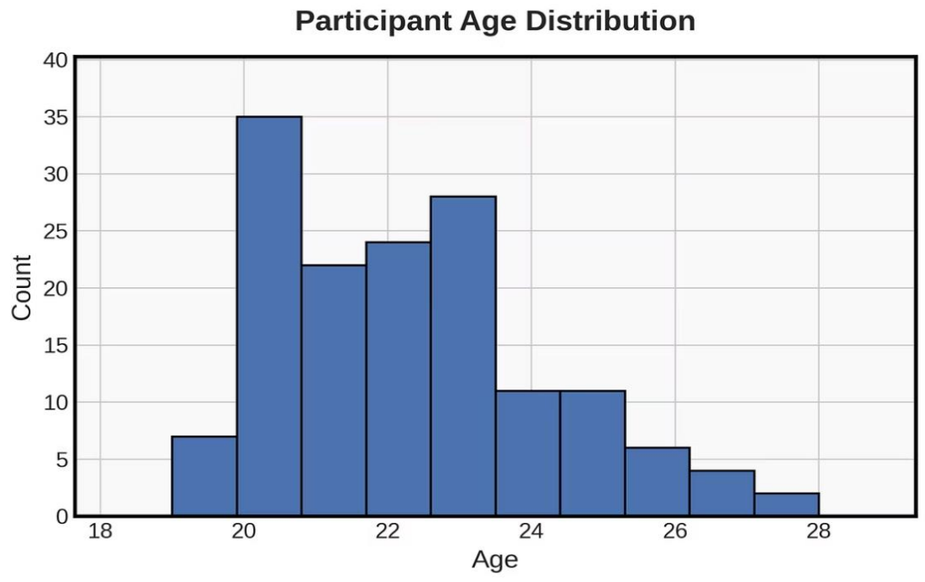
Frames from the middle of the event

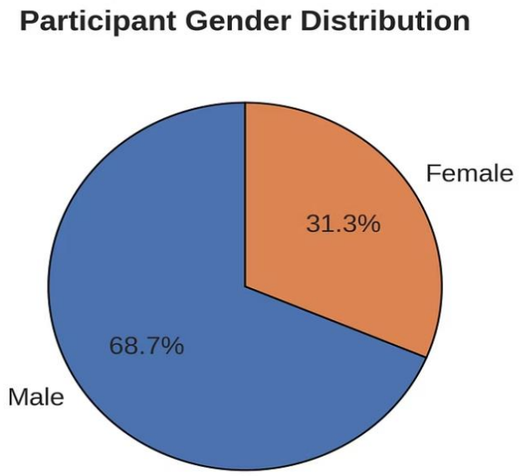
Data Cleaning & Exclusions

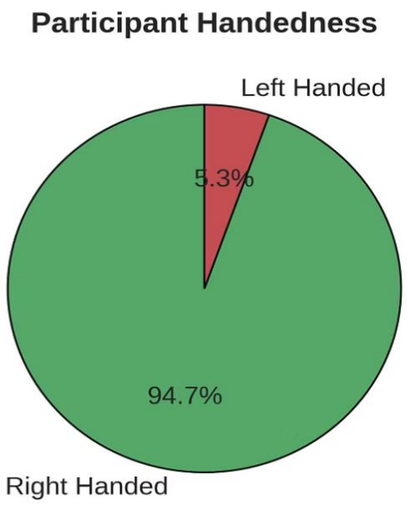
Reason	Excluded	Remaining
First 13 participants (tech error)	13	171
Incomplete data	1	170
Vigilance failures (>27.05 min)	4	**166**

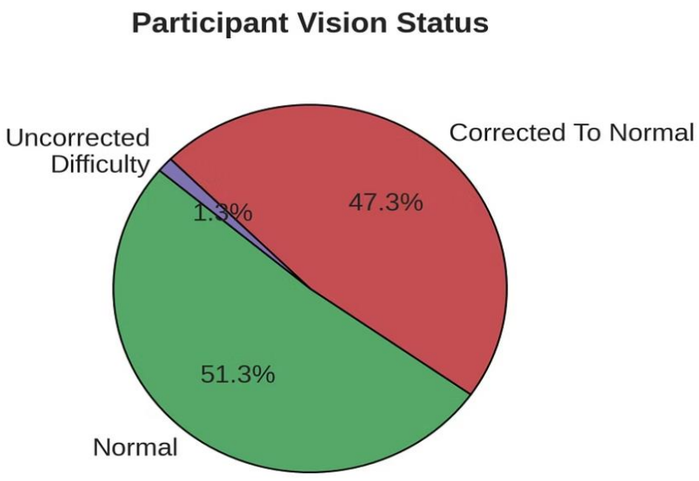
Condition	Before	Excluded	After
Abrupt Cut (AB)	81	2	**79**
Natural Cut (NB)	90	3	**87**
Total	**171**	**5**	**166**

Demographic Summary









Research Hypotheses

Five hypotheses were tested:

1. Overall accuracy and response time differences
2. Accuracy across frame types
3. Confidence differences for correct and incorrect responses
4. Confidence differences across frame categories
5. Effects of demographic factors

Hypotheses Overview

#	Research Question	IV	DV	Test
H1	Cut type → accuracy & RT?	Cut type	Accuracy, RT	t-test
H2	Cut type → accuracy by frame type?	Cut type	Accuracy (BB/EM)	t-test
H3	Cut type → confidence by correctness?	Cut type	Confidence	t-test
H4	Cut type → confidence by frame type?	Cut type	Confidence (BB/EM)	t-test
H5	Demographics → accuracy?	Age, Gender, etc.	Accuracy	t-test / correlation

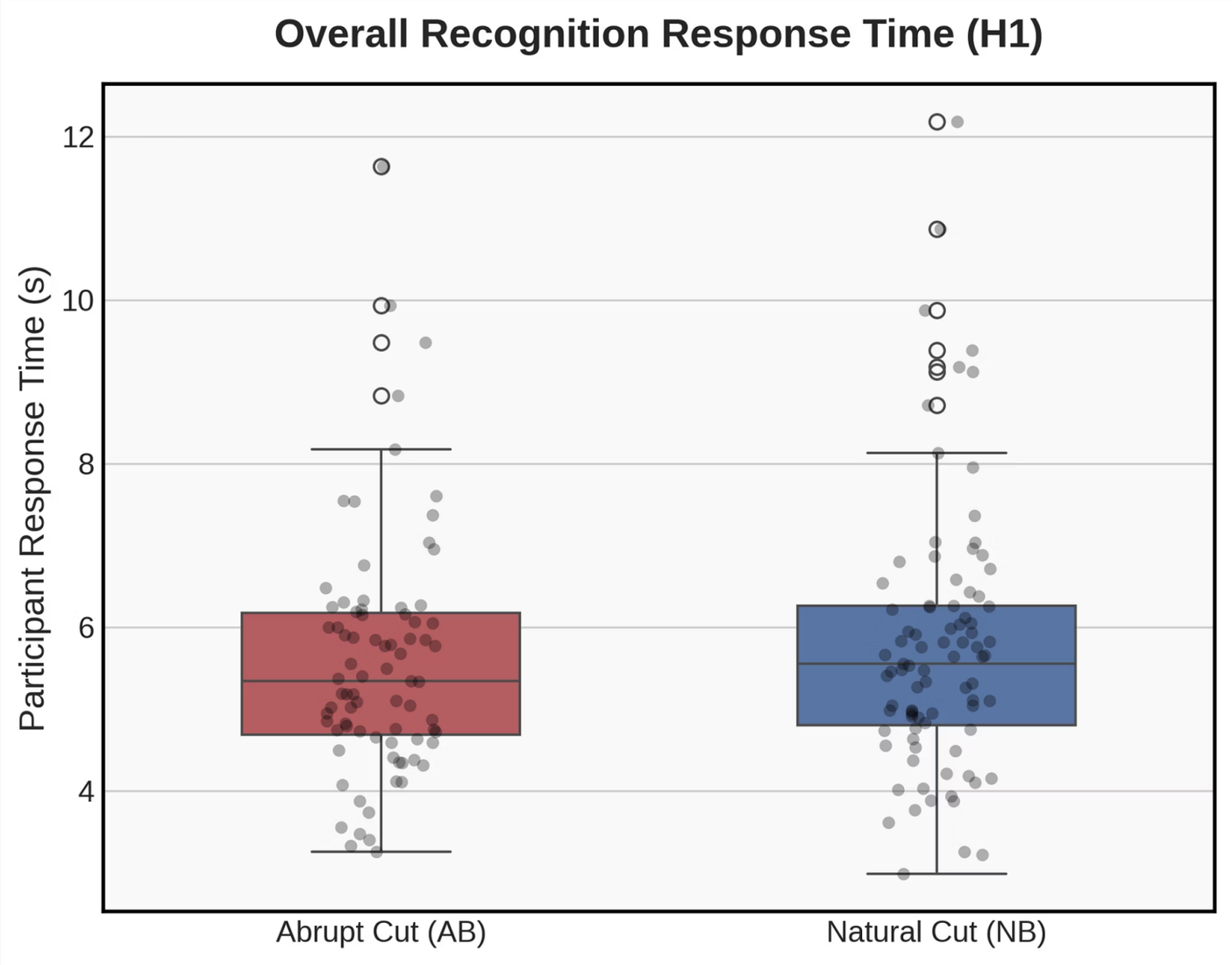
Hypothesis – 1 : Overall Recognition Accuracy (Statistical Test)

H_0: No difference in accuracy between NB and AB groups
H_1: NB group has higher accuracy than AB group

Group	n	M	SD	IQR
Natural Cut (NB)	87	0.870	0.073	0.825–0.925
Abrupt Cut (AB)	79	0.838	0.082	0.775–0.900

Result: t = 2.707, p = 0.004 (**) → Reject H_0 NB group had significantly higher accuracy.

Hypothesis – 1 : Overall Recognition Accuracy (Data Visualization)



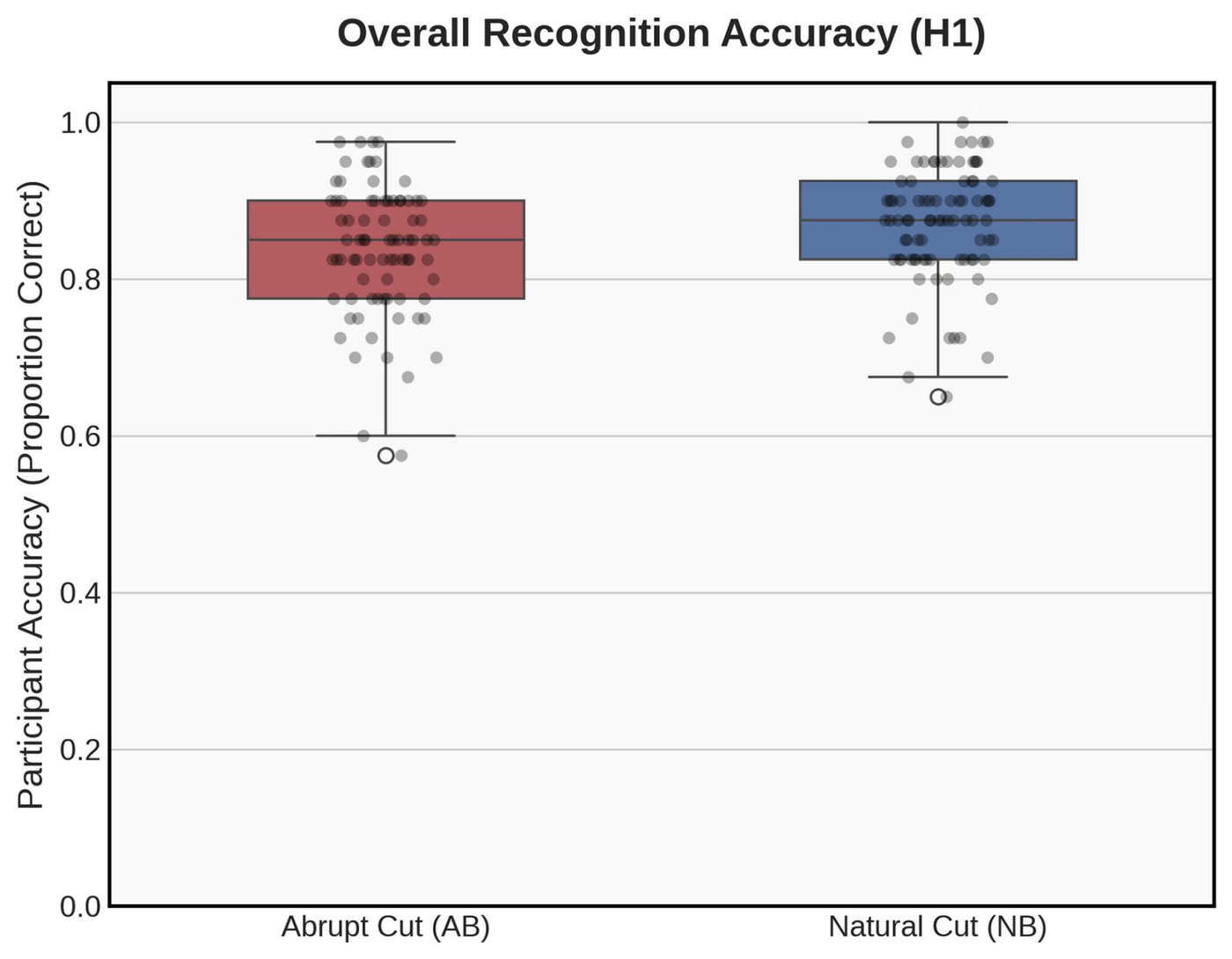
Hypothesis – 1 : Overall Response Times (Statistical Test)

H_0: No difference in RT between NB and AB groups
H_1: NB group responds faster than AB group

Group	n	M (s)	SD	IQR
Natural Cut (NB)	87	5.765	1.637	4.800–6.261
Abrupt Cut (AB)	79	5.581	1.484	4.688–6.174

Result: t = 0.761, p = 0.776 (ns) → Fail to reject H_0 No significant difference in response times.

Hypothesis – 1 : Overall Response Time (Data Visualization)



Hypothesis – 2 : Accuracy Across Frame Types (Statistical Test)

Part A: Before Boundary (BB)

H_0: No difference in accuracy for BB frames
H_1: Natural Cut (NB) group has higher accuracy for BB frames

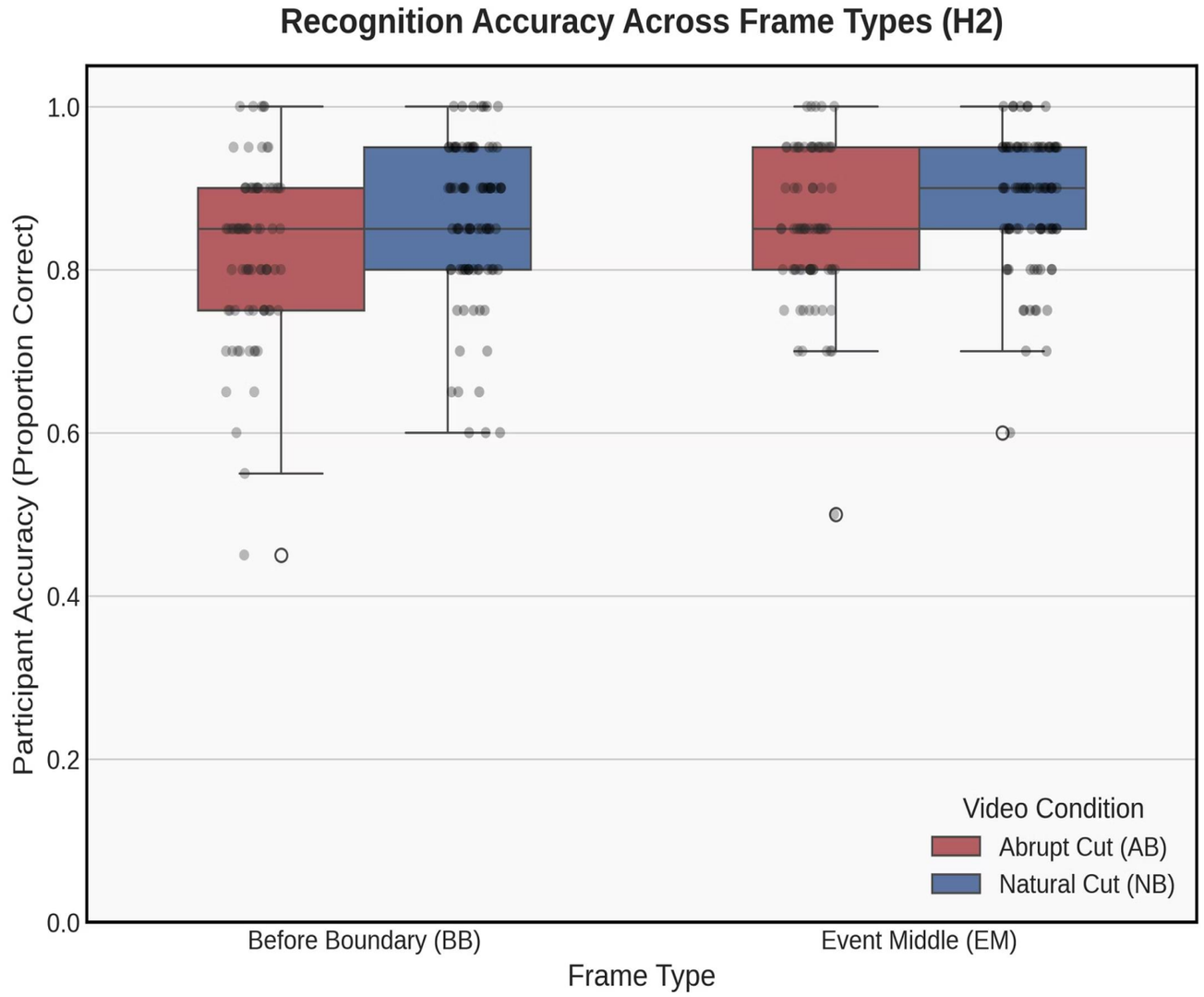
Part B: Event Middle (EM)

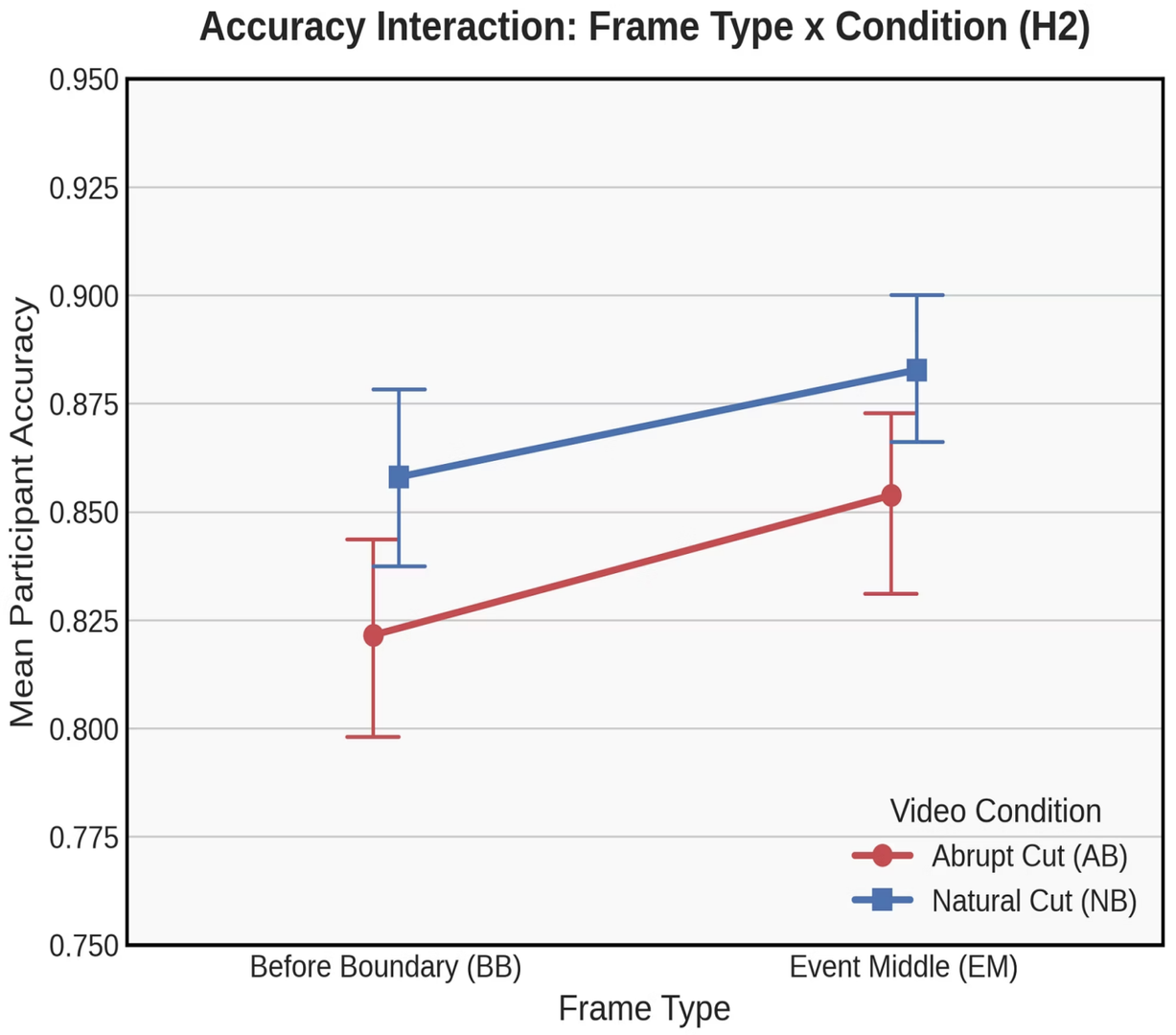
H_0: No difference in accuracy for EM frames
H_1: Accuracy for EM frames differs between groups

Condition	Frame Type	n	M	SD
NB	Before Boundary	87	0.858	0.097
AB	Before Boundary	79	0.822	0.105
NB	Event Middle	87	0.883	0.080
AB	Event Middle	79	0.854	0.093

Result (Part A): t = 2.329, p = 0.011 (*) (one-tailed)→ Reject H_0
Result (Part B): t = 2.151, p = 0.033 (*) (two-tailed)→ Reject H_0
Accuracy is significantly higher for both boundary and middle frames in the Natural Cut group compared to the Abrupt Cut group.

Hypothesis – 2 : Accuracy Across Frame Types (Data Visualization)





Hypothesis – 3 : Confidence by Correctness (Statistical Test)

Part A: Correct Responses

H_0: No difference in confidence ratings for correct responses
H_1: NB group reports higher confidence for correct responses

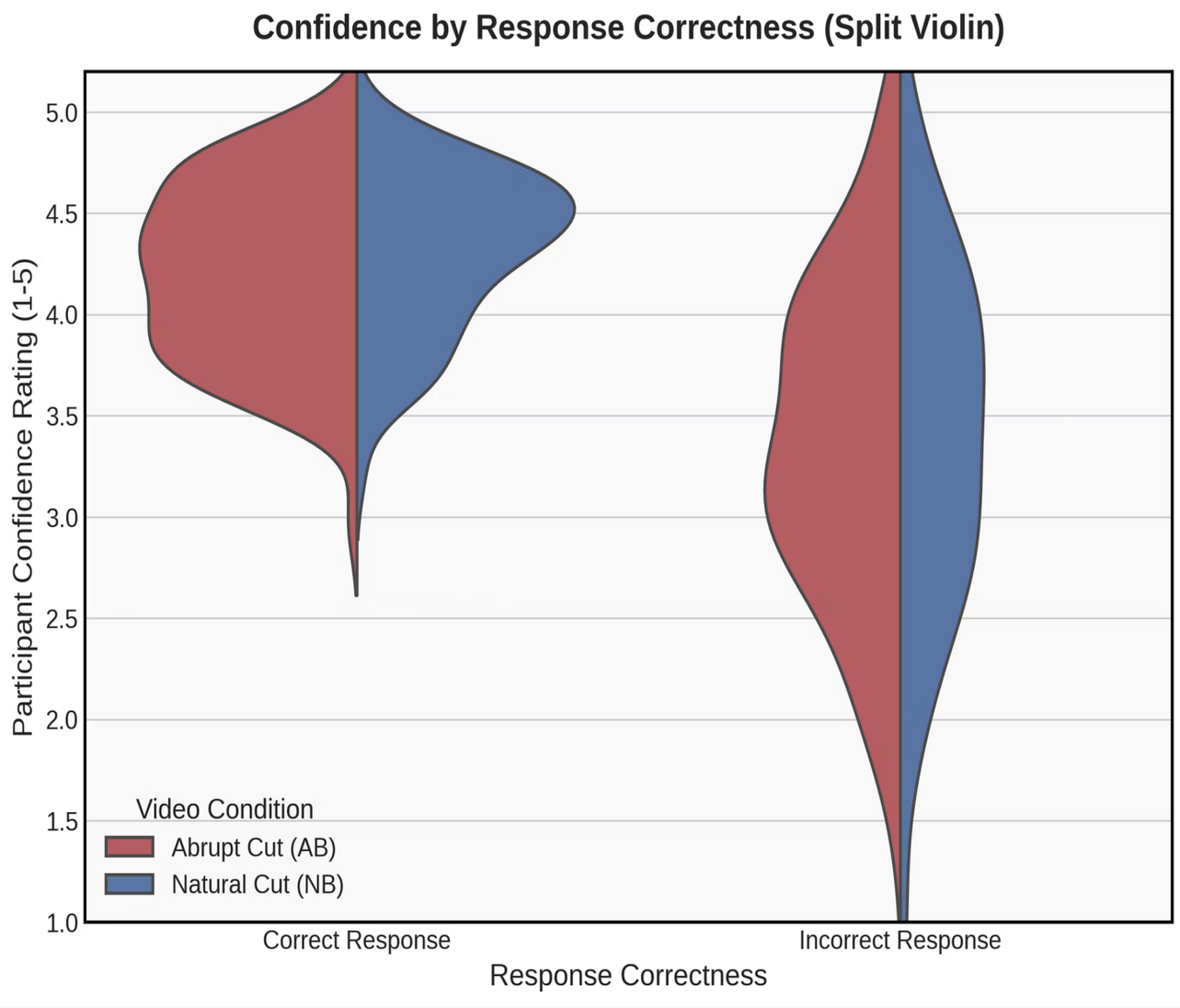
Part B: Incorrect Responses

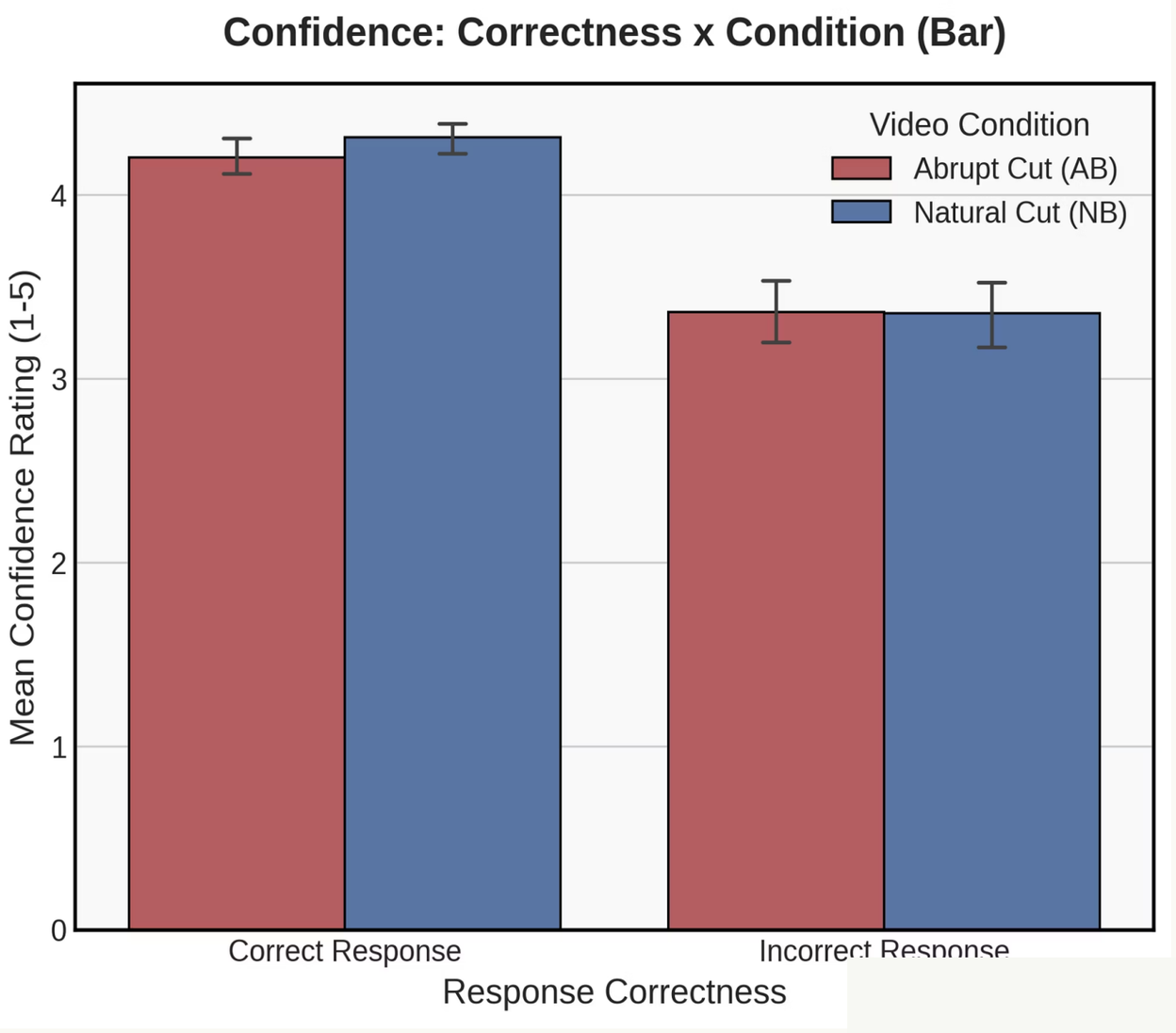
H_0: No difference in confidence ratings for incorrect response
H_1: Confidence for incorrect responses differs between groups

Condition	Correctness	n	M	SD
NB	Correct	87	4.314	0.389
AB	Correct	79	4.204	0.430
NB	Incorrect	86	3.356	0.874
AB	Incorrect	79	3.362	0.787

Result (Part A): t = 1.717, p = 0.044 (*) (one-tailed) → Reject H_0
Result (Part B): t = -0.041, p = 0.968 (ns) (two-tailed) → Fail to reject H_0
The Natural Cut group reported significantly higher confidence for correct responses, but there was no significant difference in confidence for incorrect responses between groups.

Hypothesis – 3 : Confidence by Correctness (Data Visualization)





Hypothesis – 4 : Confidence by Frame Type (Statistical Test)

Part A: Before Boundary (BB)

H_0: No difference in Confidence for BB frames
H_1: AB group reports lower confidence for boundary frames

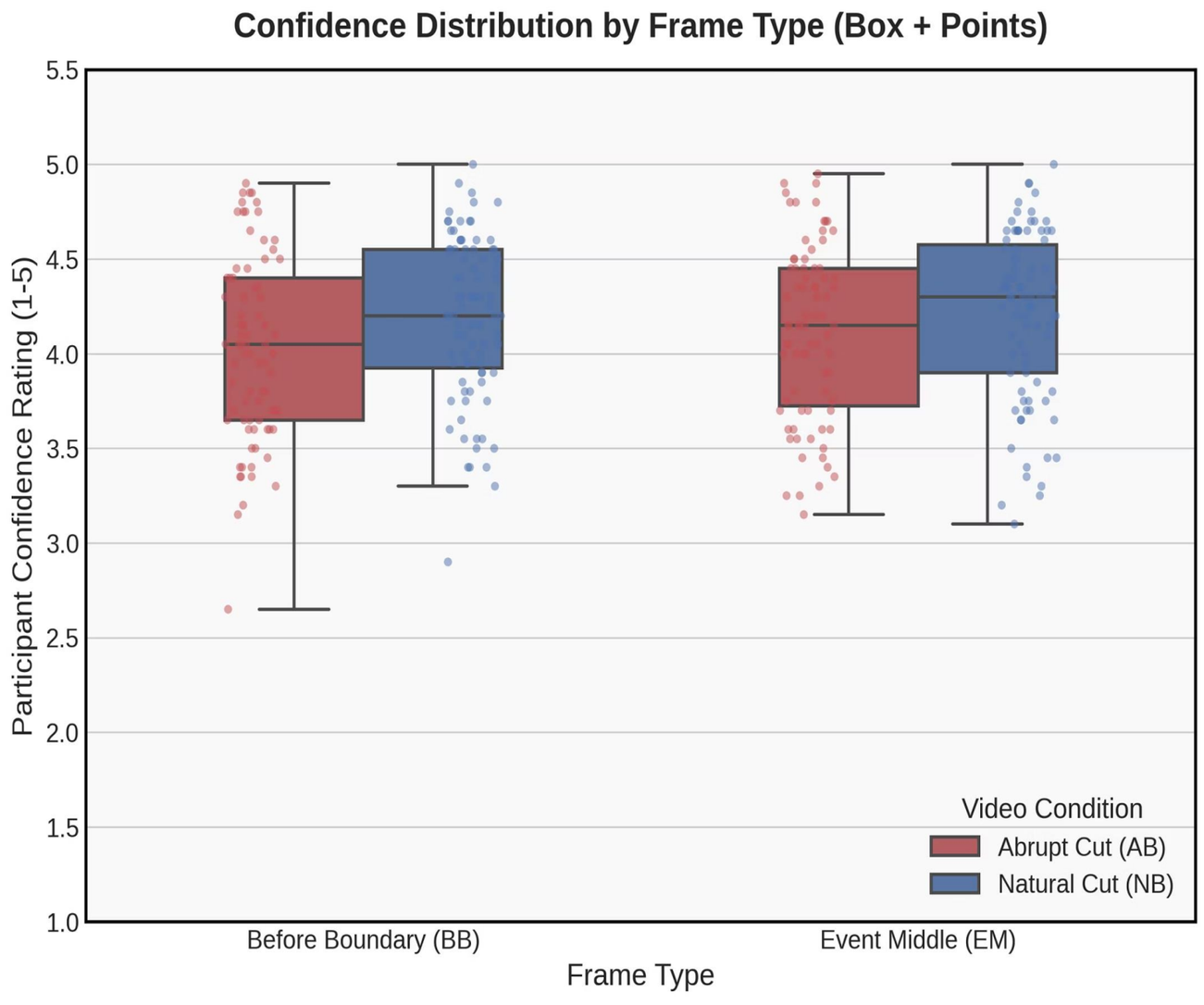
Part B: Event Middle (EM)

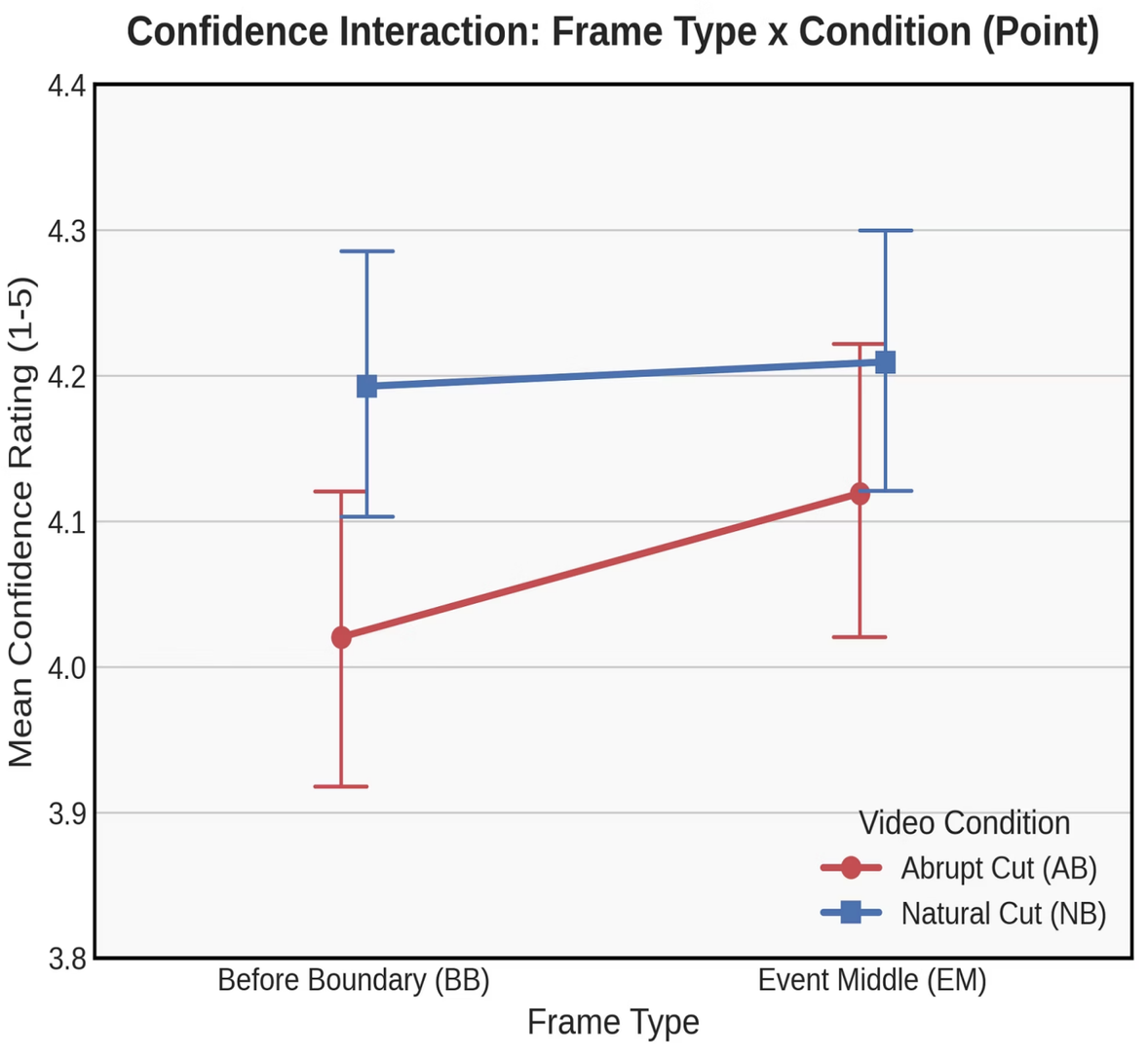
H_0: No difference in Confidence for EM frames
H_1: Confidence for EM frames differs between groups

Condition	Frame Type	n	M	SD
NB Before	Boundary	87	4.193	0.428
AB Before	Boundary	79	4.020	0.491
NB Event	Middle	87	4.209	0.446
AB Event	Middle	79	4.119	0.461

Result (Part A): t = 2.399, p = 0.009 (*) (one-tailed) → Reject H_0
Result (Part B): t = 1.279, p = 0.203 (ns) (two-tailed) → Fail to reject H_0
The confidence drop was specific to boundary frames in the Abrupt Cut group, with no significant difference for middle frames.

Hypothesis – 4 : Confidence by Frame Type (Data Visualization)





Hypothesis – 5 : Demographic Factor (Statistical Test)

H_0: Demographic factors have no significant effect on accuracy

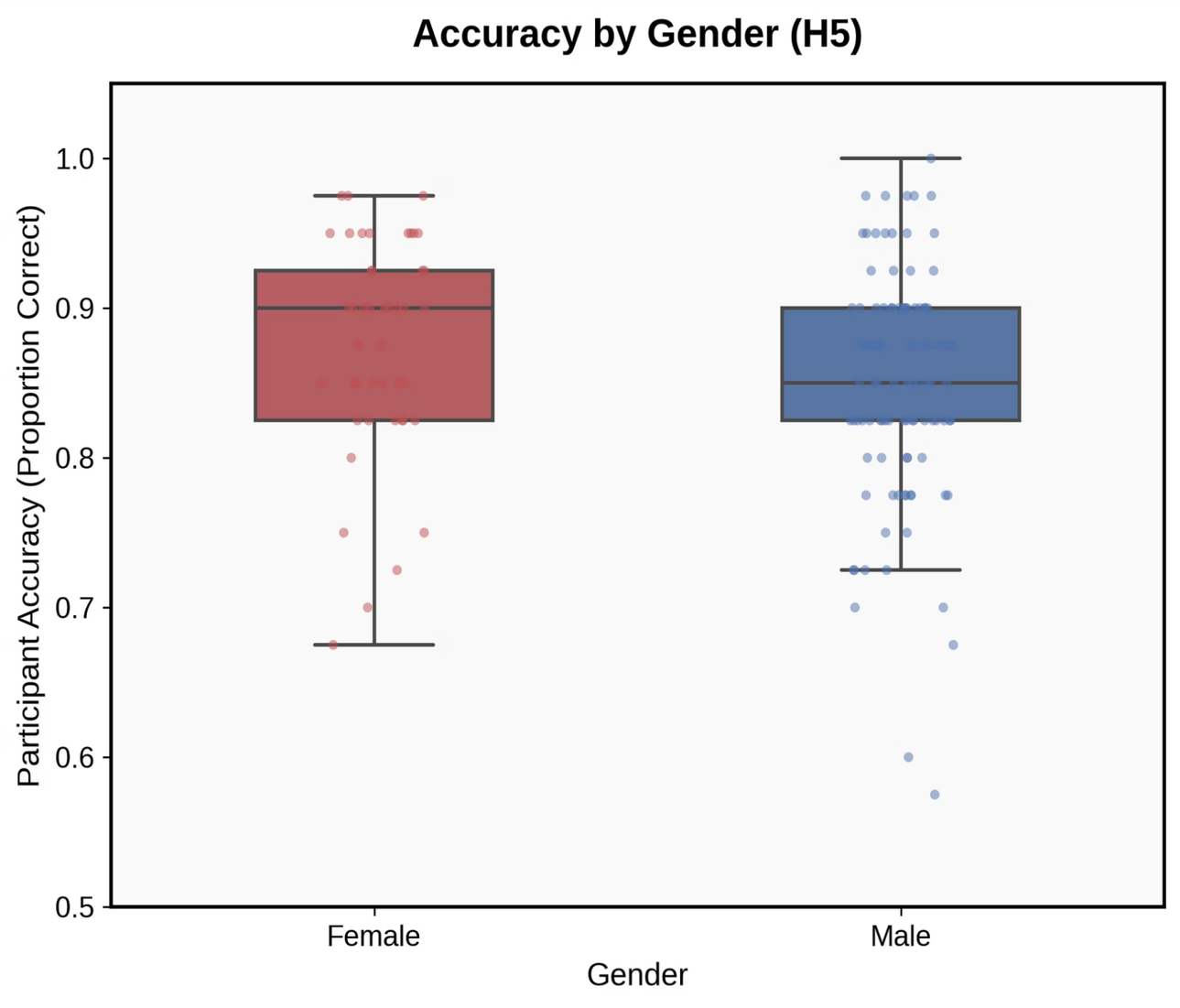
H_1: Demographic factors significantly affect recognition accuracy

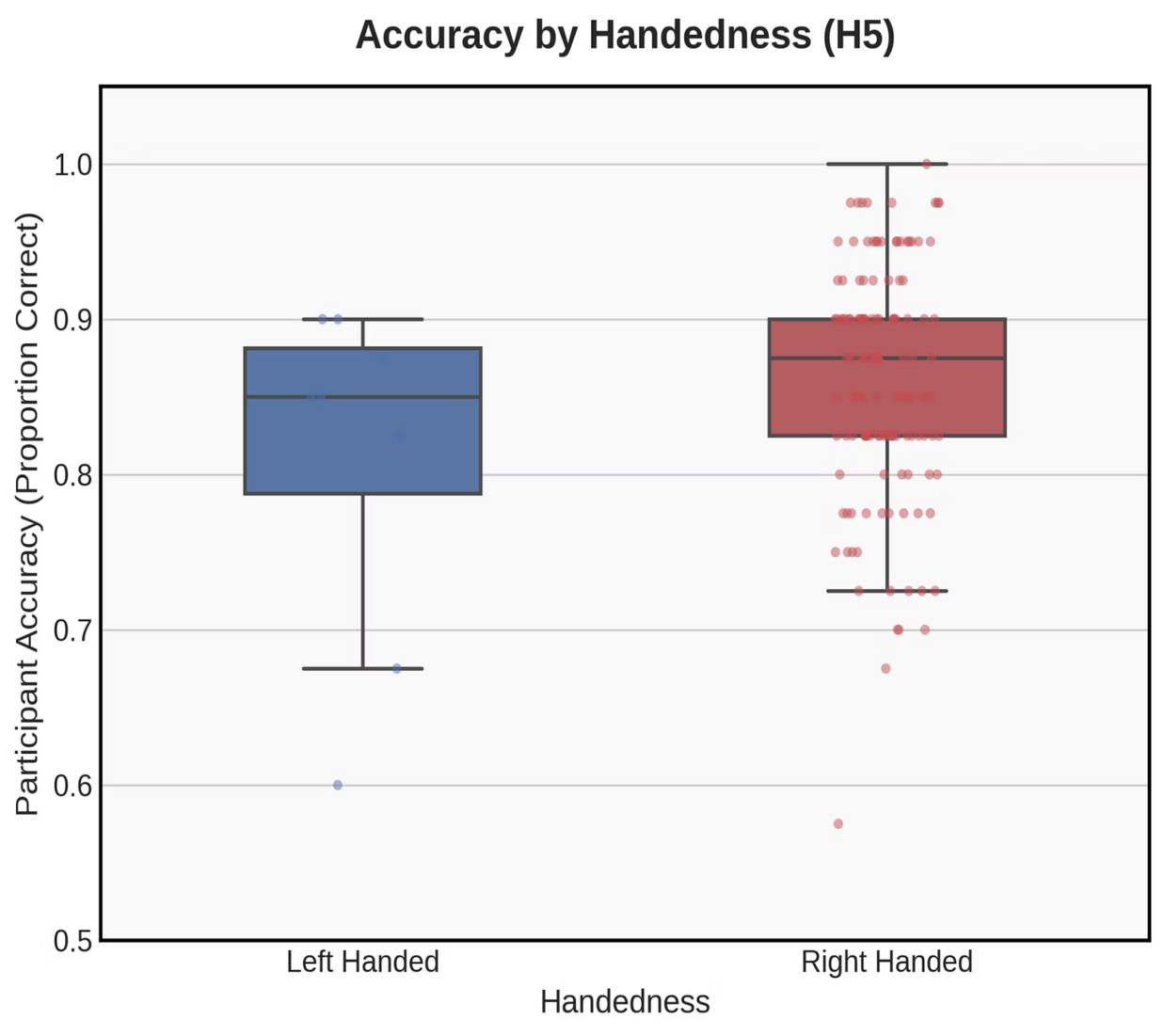
Group	n	M	SD
Male	103	0.849	0.074
Female	45	0.875	0.073
Right-Handed	140	0.860	0.076
Left-Handed	8	0.809	0.092
Normal Vision	75	0.856	0.076
Corrected Vision	71	0.855	0.077

Test	Statistic	p	Sig
Male vs Female	t = -1.956	0.054	ns
Right vs Left-Handed	t = 1.261	0.246	ns
Accuracy ~ Age	r = -0.147	0.075	ns
Normal vs Corrected Vision	t = 0.111	0.911	ns

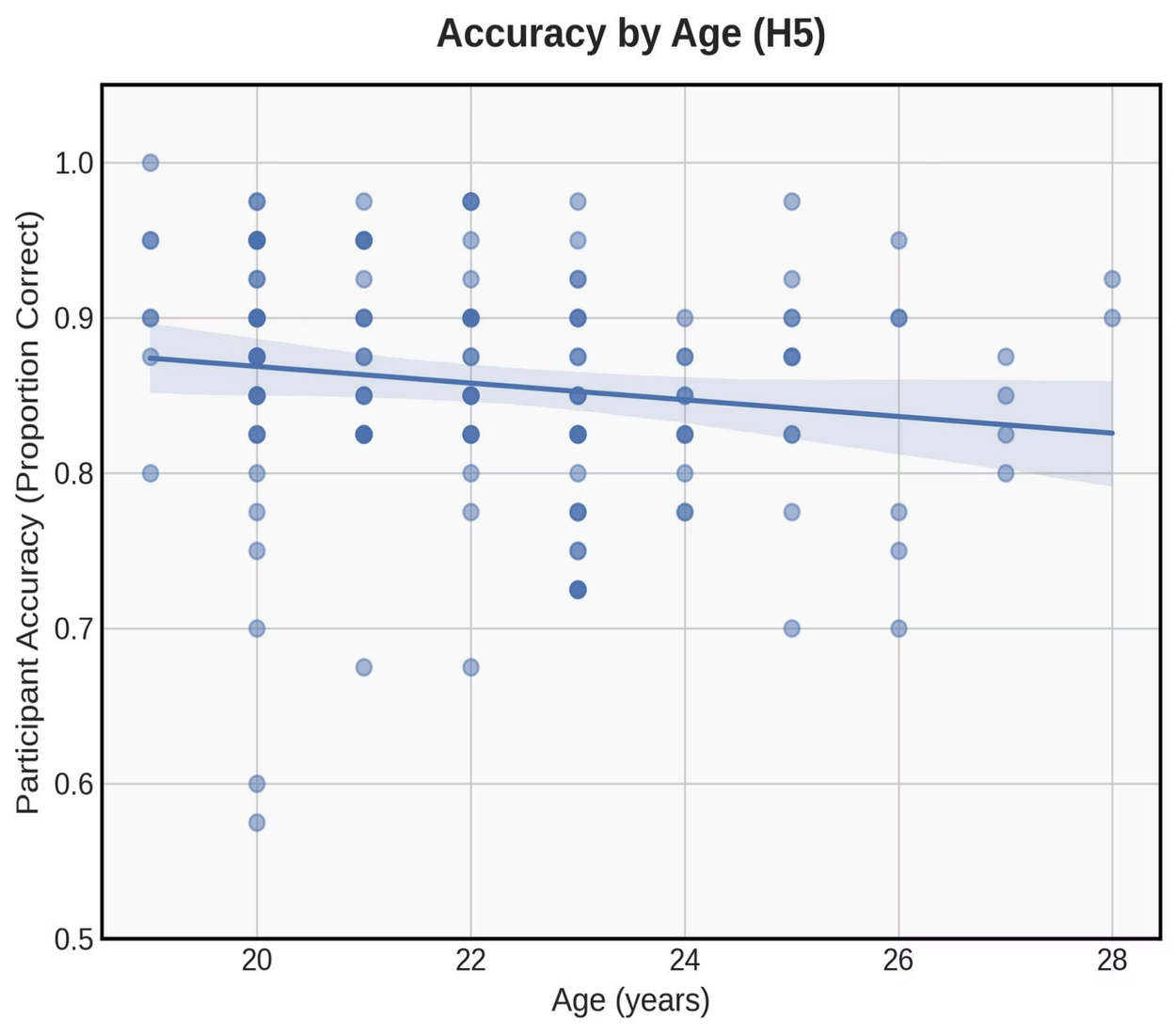
Result: All p-values > 0.05 → Fail to reject H_0 No demographic factor significantly affected recognition accuracy.

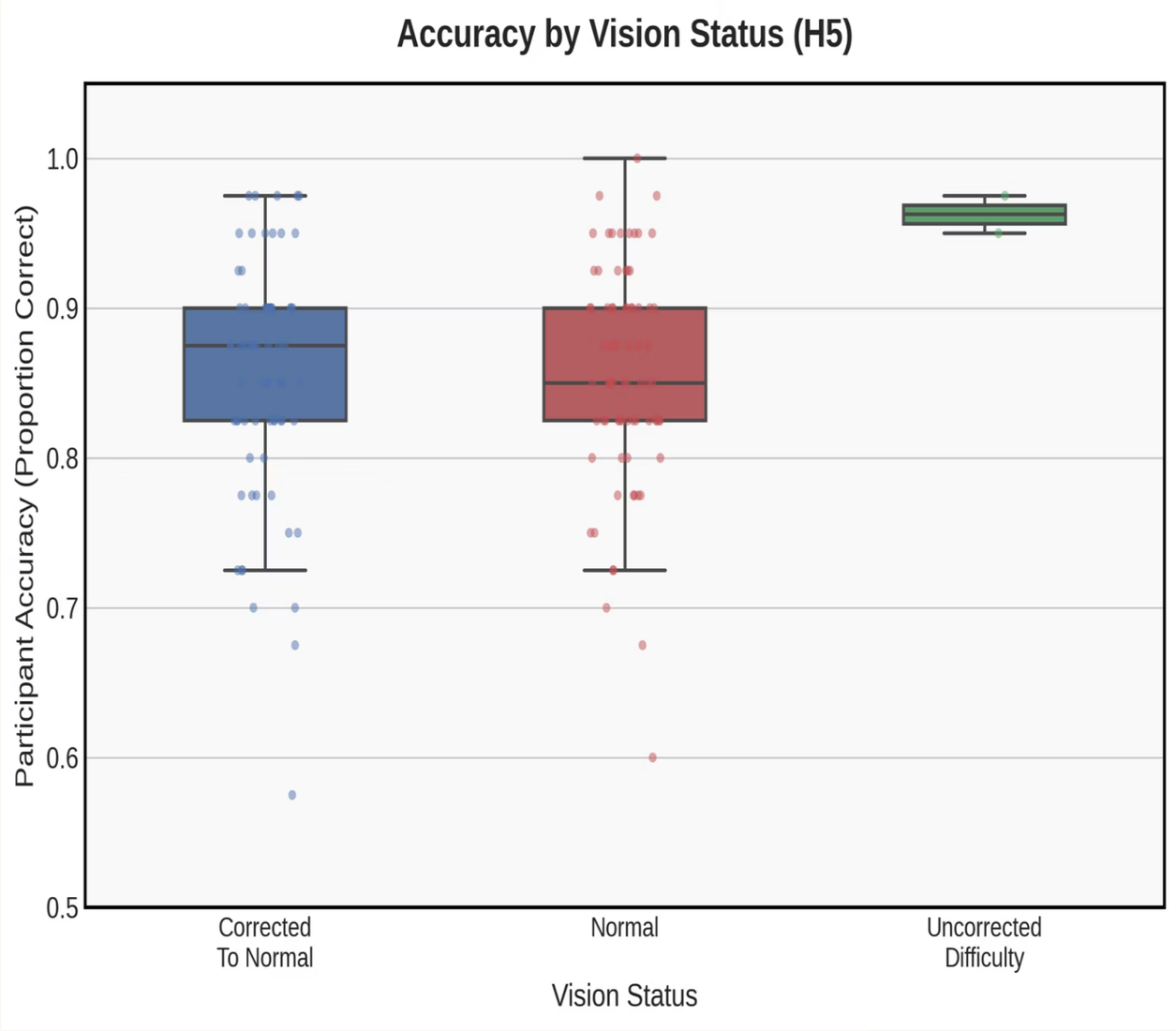
Hypothesis – 5 : Demographic Factor (Data Visualization)





Hypothesis – 5 : Demographic Factor (Data Visualization)





Summary of Results

Hypothesis	Finding	p-value	Decision	
H1: Accuracy NB > AB	NB significantly higher	**0.004**	Reject H_0	✓
H1: RT NB < AB	No difference	0.776	Fail to reject H_0	
H2: BB Accuracy NB > AB	NB significantly higher	**0.011**	Reject H_0	✓
H2: EM Accuracy NB vs AB	NB significantly higher	**0.033**	Reject H_0	✓
H3: Confidence (Correct) NB > AB	NB significantly higher	**0.044**	Reject H_0	✓
H3: Confidence (Incorrect) NB vs AB	No difference	0.968	Fail to reject H_0	
H4: Confidence BB NB > AB	NB significantly higher	**0.009**	Reject H_0	✓
H4: Confidence EM NB vs AB	No difference	0.203	Fail to reject H_0	
H5: Demographics → Accuracy	No significant effects	>0.05	Fail to reject H_0	

Key Takeaways

1. Abrupt cuts impair memory

NB group had ~3.2% higher recognition accuracy ($p = 0.004$)

2. Both frame types affected

Not just boundary frames, but also mid-event frames

3. Confidence drops for boundary frames

AB group was less confident specifically for frames near the disrupted cut ($p = 0.009$)

4. Correct answers come with more confidence in NB

But no difference when wrong

5. Response time is unaffected

The cut doesn't slow people down, it just makes them less accurate

6. Demographics don't matter

Age, gender, handedness, and vision had no significant effect

Future Directions

Three promising directions for expanding our understanding of how abrupt cuts impact memory storage and recall in the perception of films.

01

Signal Detection Theory Analysis

Apply d’ measures to determine whether sudden cuts cause response bias in recognition judgments or result in weaker memory traces.

02

Video Duration as Moderator

Check whether longer video context reduces the effects of disruption.
Longer clips may reveal the occurrence's boundary conditions and provide richer semantic structure that protects memory from cut-related interference

03

Unified ANOVA Framework

Examine interaction effects of cut type × frame type within a single comprehensive model. This method will show whether boundary frames are more sensitive to sudden changes than to gradual ones.

References

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Thank You